Validation and Calibration of the Remote Consolidated Approach of Reporting on Food Security (rCARI) Methodology

Preliminary results from Central African Republic

Version 1.0

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1 Introduction

The overall goal of this study was to assess the validity of the rCARI module(s) against conventional CARI modules for measuring food security within the same population. The study compares indicators collected through phone surveys with conventional face-to-face survey, considering the latter as the standard standard methodology/gold standard. The following food security indicators were compared: Food Consumption Score (FCS), reduced Coping Strategy Index (rCSI), Livelihood Coping Strategy (LCS), Food Expenditure Share (FES), and income sources and income change. The Economic Capacity to Meet Essetial Needs (ECMEN) indicator which is sometimes used to aggregate CARI will not be tested in this study, as it requires a standard long expenditure module which cannot be collected over the phone. In addition, ECMEN calculation requires having an updated value of the minimum expenditure basket (MEB) of the study population, which is not available for the study areas. In its stead, a shortened version of the standard expenditure module will be used to test the possibility of collecting FES data over the phone.

Studying the validity of using the phone to collect food security indicators is a key step in judging the appropriateness of the rCARI methodology to report food insecurity prevalence. However, Testing the comparability of these indicators through different modalities of data collection (Face-to-Face (F2F) vs. Phone) should be read within the wider consideration around phone survey modality of data collection. For example, other factors aside the modules should be considered while reading results of any remote survey.

1 First, mobile phone ownership (mobile phone penetration rate in the study area) could likely exclude the most vulnerable population from the sample.

2 Second, mobile phone network coverage; Even in countries with high penetration rates, network coverage could be a problem, especially when electricity cuts are frequent and local infrastructure, such as communication masts, are not evenly distributed (World Bank, 2020).

3 Third, response rate: It is well documented that the response rate in phone surveys is usually lower than in face-to-face surveys (L'Engle et al., 2018; World Bank, 2020). Response rate is usually lower among the well-off population which could contribute to sample skewness.

These factors may lead to skewed sample from phone surveys especially if the goal to report national representative results/outcomes. Nevertheless, there are several statistical weighting techniques that can adjust the results of phone surveys based baseline variables like using demographics characteristics of the most recent face-to-face surveys (Nagpal et al. 2021; Glazerman et al. 2020; Valliant et al. 2013; Elkasabi, 2015). This allows the final results of such exercises to be aggregated and extrapolated to reflect some level of representativeness. However, this current study remains insulated from these confounding issues since the main idea is not to report results at the national level but rather to compare how the two methods perform within the same population with similar characteristics. For this reason, the study does not focus on weighting techniques to adjust for sampling skewness and response rate. Nonetheless, the study design adopted in Central

African Republic (CAR), could allow for testing of sample skewness due to mobile phone ownership and coverage (see CAR sample design in Annex III). CAR has one of the lowest mobile phone penetration rates globally (approximately 70% of the population do not own/use mobile phones).

2 Study are and sample selection

The selection of households for the listing followed a stratified random approach selection based on 3-point criteria developed and agreed together with the country office in Bangui.

1. Mobile phone coverage
2. Food security situation
3. Accessibility (considering the security situation on the ground)

Two Administration 1 (ADMIN1) Areas were selected based on the afore mentioned criteria.

1. Nana- Gribizi to the West of the country and,
2. Mbomou to the East of the country

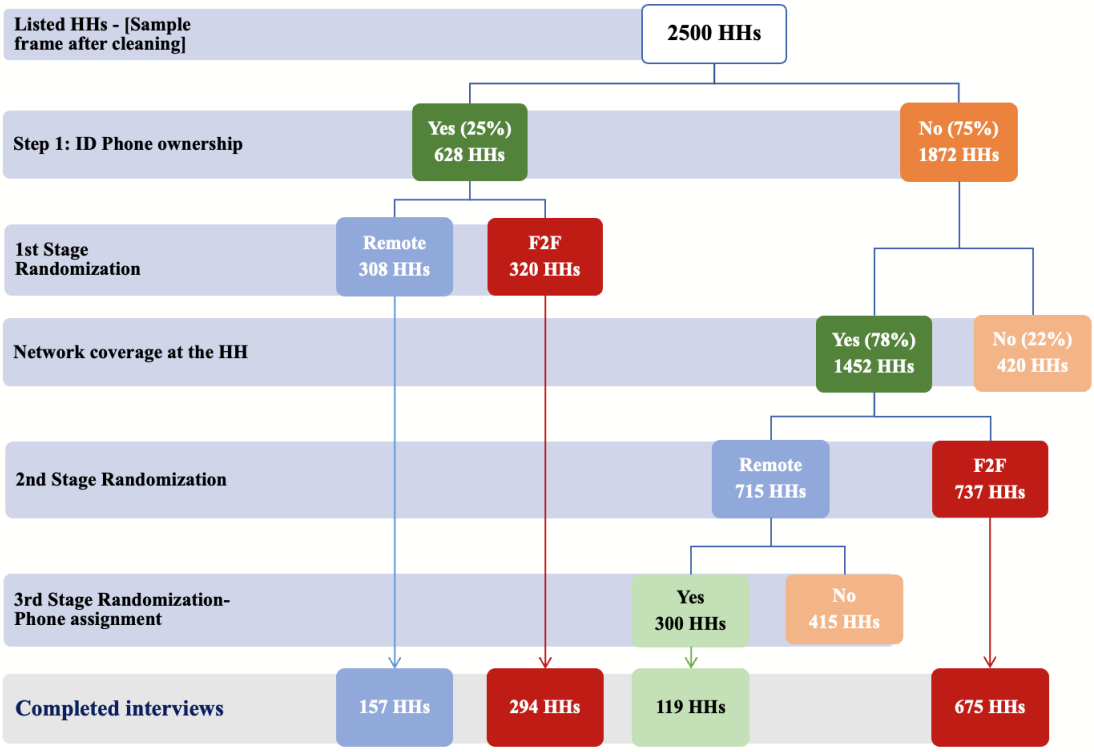
Selection of Admin2, 3 and 5 were randomly done based on the compiled list of areas that had mobile phone connectivity-The full list of areas with connectivity has been compiled by Orange/Moove/Telecel the biggest mobile phone operators in the country. This is to ensure that households sampled would be able to receive remote interviews if they were to be randomly assigned to the remote module.

3 Sample Completion

Table 1: Sample Completion: Face-to-Face vs. Remote

	Not Completed		Completed		Total	
	Count	Percent	Count	Percent	Count	Percent
Face-to-Face	88	8.33	969	91.67	1057	100.00
Remote	747	73.02	276	26.98	1023	100.00
Total	835	40.14	1245	59.86	2080	100.00

Figure 1: Sample Composition and Completion in CAR



4 Randomization Quality

To assess the balance of the treatment (Remote) and control (F2F) groups, we examined various respondent and household characteristics, including basic demographics, income sources, and changes over time. This allowed us to evaluate whether the two groups were similar in terms of these factors, ensuring a fair comparison between the treatment and control conditions.

4.1 Modality Randomization

Table 2: Balance Table on Randomized Survey Modality Assignment: Face-to-Face vs. Remote

	Own Phone			No Phone			Total		
Variable	F2F	Remote	Diff	F2F	Remote	Diff	F2F	Remote	Diff
List respondent age	35.97	37.59	1.61	39.42	39.35	-0.08	38.38	38.82	0.44
	(12.42)	(12.76)	(0.11)	(13.95)	(14.24)	(0.92)	(13.59)	(13.83)	(0.47)
Household head is the respondent	0.72	0.73	0.01	0.78	0.79	0.02	0.76	0.78	0.02
	(0.45)	(0.44)	(0.68)	(0.42)	(0.40)	(0.43)	(0.43)	(0.42)	(0.38)
Female Household Head	0.20	0.23	0.03	0.30	0.31	0.01	0.27	0.28	0.02
	(0.40)	(0.42)	(0.30)	(0.46)	(0.46)	(0.66)	(0.44)	(0.45)	(0.37)
Married Household Head	0.25	0.26	0.01	0.14	0.11	-0.03	0.17	0.16	-0.02
	(0.44)	(0.44)	(0.85)	(0.35)	(0.31)	(0.09)	(0.38)	(0.36)	(0.25)
Household head years of education	7.50	7.49	-0.02	3.39	3.30	-0.08	4.63	4.56	-0.07
	(4.10)	(4.50)	(0.96)	(3.27)	(3.22)	(0.63)	(4.01)	(4.12)	(0.70)
Displaced household	0.11	0.12	0.01	0.03	0.03	-0.00	0.06	0.06	0.00
	(0.31)	(0.32)	(0.67)	(0.18)	(0.18)	(0.85)	(0.23)	(0.23)	(0.86)
Household size	7.68	7.95	0.28	6.09	5.96	-0.13	6.57	6.56	-0.01
	(3.50)	(3.95)	(0.35)	(2.66)	(2.65)	(0.37)	(3.03)	(3.23)	(0.96)
Number of household member younger than 15	4.02	4.00	-0.02	3.21	3.13	-0.08	3.45	3.39	-0.06
	(2.27)	(2.12)	(0.93)	(1.92)	(1.90)	(0.41)	(2.06)	(2.01)	(0.47)
Number of household member older than 60	0.21	0.22	0.01	0.18	0.18	0.00	0.19	0.19	0.01
	(0.48)	(0.55)	(0.72)	(0.44)	(0.48)	(0.87)	(0.45)	(0.50)	(0.73)
Has household member with disability/chronic disease	0.29	0.30	0.00	0.15	0.16	0.01	0.19	0.20	0.01
	(0.46)	(0.46)	(0.96)	(0.36)	(0.37)	(0.50)	(0.40)	(0.40)	(0.59)
Number of rooms	2.84	2.99	0.15	1.96	1.91	-0.05	2.23	2.24	0.01
	(1.25)	(1.34)	(0.14)	(1.08)	(1.09)	(0.40)	(1.20)	(1.27)	(0.83)
Household Crowding Index	3.08	3.00	-0.08	4.02	4.00	-0.02	3.74	3.70	-0.03
	(1.79)	(1.64)	(0.57)	(2.94)	(2.97)	(0.92)	(2.68)	(2.68)	(0.78)
Household has impoverished roof	0.67	0.66	-0.01	0.85	0.86	0.01	0.79	0.80	0.00
	(0.47)	(0.48)	(0.80)	(0.36)	(0.35)	(0.57)	(0.41)	(0.40)	(0.79)
Household has impoverished floor	0.89	0.91	0.02	0.96	0.96	0.00	0.94	0.95	0.01
	(0.32)	(0.29)	(0.37)	(0.19)	(0.19)	(0.87)	(0.24)	(0.22)	(0.44)
Observations	320	308	628	737	715	1,452	1,057	1,023	2,080

Note: Robust standard errors in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

4.2 F2F Length Randomization

Table 3: Balance Table on Randomized Survey Length Assignment in F2F Survey: Long vs. Short

	Own Phone			No Phone			Total F2F		
Variable	Long	Short	Diff	Long	Short	Diff	Long	Short	Diff
List respondent age	35.72	36.24	0.52	40.13	38.66	-1.47	38.81	37.92	-0.89
	(12.73)	(12.11)	(0.71)	(14.66)	(13.11)	(0.15)	(14.24)	(12.85)	(0.29)
Household head is the respondent	0.71	0.72	0.00	0.80	0.76	-0.04	0.77	0.75	-0.03
	(0.45)	(0.45)	(0.93)	(0.40)	(0.43)	(0.22)	(0.42)	(0.44)	(0.33)
Female Household Head	0.20	0.19	-0.01	0.30	0.30	-0.00	0.27	0.26	-0.01
	(0.40)	(0.40)	(0.84)	(0.46)	(0.46)	(0.94)	(0.44)	(0.44)	(0.85)
Married Household Head	0.24	0.26	0.02	0.12	0.16	0.03	0.16	0.19	0.03
	(0.43)	(0.44)	(0.70)	(0.33)	(0.37)	(0.18)	(0.37)	(0.39)	(0.19)
Household head years of education	7.30	7.71	0.41	3.32	3.46	0.13	4.52	4.76	0.24
	(4.04)	(4.16)	(0.38)	(3.23)	(3.32)	(0.58)	(3.94)	(4.09)	(0.34)
Displaced household	0.13	0.08	-0.06	0.04	0.02	-0.02	0.07	0.04	-0.03*
	(0.34)	(0.27)	(0.10)	(0.21)	(0.15)	(0.10)	(0.26)	(0.19)	(0.02)
Household size	7.46	7.91	0.45	6.17	6.00	-0.17	6.56	6.58	0.03
	(3.35)	(3.65)	(0.25)	(2.82)	(2.47)	(0.38)	(3.05)	(3.01)	(0.89)
Number of household member younger than 15	3.87	4.17	0.30	3.28	3.13	-0.15	3.46	3.45	-0.01
	(2.13)	(2.41)	(0.24)	(2.07)	(1.74)	(0.29)	(2.10)	(2.02)	(0.95)
Number of household member older than 60	0.21	0.21	0.00	0.19	0.16	-0.03	0.20	0.17	-0.02
	(0.49)	(0.47)	(0.94)	(0.43)	(0.45)	(0.30)	(0.45)	(0.45)	(0.43)
Has household member with disability/chronic disease	0.32	0.27	-0.05	0.16	0.14	-0.01	0.21	0.18	-0.02
	(0.47)	(0.44)	(0.35)	(0.36)	(0.35)	(0.61)	(0.40)	(0.39)	(0.34)
Number of rooms	2.78	2.90	0.12	2.02	1.89	-0.13	2.25	2.20	-0.05
	(1.25)	(1.24)	(0.38)	(1.10)	(1.05)	(0.10)	(1.20)	(1.20)	(0.50)
Household Crowding Index	3.04	3.13	0.08	3.88	4.17	0.29	3.63	3.85	0.22
	(1.71)	(1.89)	(0.67)	(2.76)	(3.13)	(0.18)	(2.52)	(2.85)	(0.18)
Household has impoverished roof	0.70	0.63	-0.06	0.84	0.86	0.02	0.79	0.79	-0.00
	(0.46)	(0.48)	(0.25)	(0.37)	(0.35)	(0.36)	(0.41)	(0.41)	(0.92)
Household has impoverished floor	0.92	0.85	-0.07	0.96	0.96	0.00	0.95	0.93	-0.02
	(0.27)	(0.36)	(0.06)	(0.19)	(0.19)	(0.85)	(0.22)	(0.26)	(0.19)
Observations	164	156	320	382	355	737	546	511	1,057

Note: Robust standard errors in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

4.3 Remote Length Randomization

Table 4: Balance Table on Randomized Survey Length Assignment in Remote Survey: Long vs. Short

	Own Phone			No Phone			Total Remote		
Variable	Long	Short	Diff	Long	Short	Diff	Long	Short	Diff
List respondent age	37.14	38.07	0.93	39.42	39.28	-0.14	38.71	38.93	0.22
	(13.19)	(12.29)	(0.52)	(14.01)	(14.48)	(0.90)	(13.79)	(13.88)	(0.80)
Household head is the respondent	0.75	0.71	-0.03	0.81	0.78	-0.03	0.79	0.76	-0.03
	(0.44)	(0.45)	(0.54)	(0.39)	(0.42)	(0.30)	(0.41)	(0.43)	(0.25)
Female Household Head	0.20	0.27	0.07	0.31	0.31	-0.00	0.28	0.29	0.02
	(0.40)	(0.44)	(0.17)	(0.46)	(0.46)	(0.89)	(0.45)	(0.46)	(0.52)
Married Household Head	0.23	0.29	0.06	0.11	0.11	-0.00	0.15	0.16	0.01
	(0.42)	(0.46)	(0.21)	(0.32)	(0.31)	(0.92)	(0.36)	(0.37)	(0.53)
Household head years of education	7.69	7.27	-0.42	3.51	3.10	-0.41	4.81	4.31	-0.49
	(4.35)	(4.66)	(0.41)	(3.31)	(3.13)	(0.09)	(4.14)	(4.10)	(0.06)
Displaced household	0.11	0.12	0.01	0.04	0.03	-0.01	0.06	0.05	-0.01
	(0.32)	(0.33)	(0.77)	(0.19)	(0.16)	(0.29)	(0.24)	(0.23)	(0.58)
Household size	7.99	7.92	-0.07	5.91	6.02	0.11	6.55	6.57	0.02
	(3.30)	(4.57)	(0.88)	(2.79)	(2.51)	(0.58)	(3.11)	(3.36)	(0.93)
Number of household member younger than 15	3.96	4.05	0.08	3.08	3.17	0.08	3.36	3.42	0.07
	(2.01)	(2.25)	(0.73)	(2.02)	(1.76)	(0.55)	(2.06)	(1.95)	(0.59)
Number of household member older than 60	0.24	0.20	-0.04	0.14	0.22	0.07*	0.17	0.21	0.04
	(0.60)	(0.48)	(0.54)	(0.40)	(0.55)	(0.04)	(0.47)	(0.53)	(0.22)
Has household member with disability/chronic disease	0.24	0.36	0.12*	0.18	0.15	-0.04	0.20	0.21	0.01
	(0.43)	(0.48)	(0.02)	(0.39)	(0.35)	(0.19)	(0.40)	(0.41)	(0.70)
Number of rooms	3.04	2.95	-0.09	1.89	1.94	0.06	2.24	2.23	-0.01
	(1.39)	(1.29)	(0.55)	(1.10)	(1.07)	(0.49)	(1.31)	(1.23)	(0.91)
Household Crowding Index	2.95	3.06	0.11	4.07	3.94	-0.13	3.72	3.68	-0.04
	(1.29)	(1.96)	(0.56)	(3.09)	(2.84)	(0.55)	(2.71)	(2.65)	(0.81)
Household has impoverished roof	0.66	0.65	-0.01	0.85	0.86	0.01	0.79	0.80	0.01
	(0.48)	(0.48)	(0.92)	(0.36)	(0.34)	(0.68)	(0.41)	(0.40)	(0.70)
Household has impoverished floor	0.94	0.88	-0.06	0.97	0.96	-0.01	0.96	0.94	-0.02
	(0.24)	(0.33)	(0.07)	(0.18)	(0.19)	(0.68)	(0.20)	(0.24)	(0.13)
Observations	161	147	308	358	357	715	519	504	1,023

Note: Robust standard errors in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

5 Attrition Analysis

5.1 F2F Attrition

Table 5: Attrition Table on Randomized Survey Length Assignment in F2F Survey: Completed vs. Attrition

	Own Phone			No Phone			Total F2F		
Variable	Completed	Attrition	Diff	Completed	Attrition	Diff	Completed	Attrition	Diff
List respondent age	36.27	32.65	-3.61	39.47	38.89	-0.59	38.50	37.05	-1.46
	(12.55)	(10.44)	(0.09)	(13.82)	(15.33)	(0.77)	(13.52)	(14.30)	(0.36)
Household head is the respondent	0.72	0.62	-0.11	0.79	0.68	-0.11	0.77	0.66	-0.11*
	(0.45)	(0.50)	(0.27)	(0.41)	(0.47)	(0.08)	(0.42)	(0.48)	(0.04)
Female Household Head	0.19	0.27	0.08	0.30	0.31	0.01	0.26	0.30	0.03
	(0.39)	(0.45)	(0.38)	(0.46)	(0.46)	(0.87)	(0.44)	(0.46)	(0.54)
Married Household Head	0.27	0.12	-0.15*	0.14	0.10	-0.05	0.18	0.10	-0.08*
	(0.44)	(0.33)	(0.03)	(0.35)	(0.30)	(0.24)	(0.38)	(0.30)	(0.02)
Household head years of education	7.55	7.00	-0.55	3.41	3.16	-0.25	4.66	4.30	-0.37
	(4.13)	(3.77)	(0.48)	(3.25)	(3.52)	(0.59)	(4.02)	(3.98)	(0.40)
Displaced household	0.09	0.27	0.18*	0.03	0.06	0.03	0.05	0.12	0.08*
	(0.29)	(0.45)	(0.05)	(0.17)	(0.25)	(0.30)	(0.22)	(0.33)	(0.04)
Household size	7.72	7.23	-0.49	6.08	6.23	0.15	6.57	6.52	-0.05
	(3.44)	(4.18)	(0.56)	(2.67)	(2.58)	(0.66)	(3.02)	(3.15)	(0.88)
Number of household member younger than 15	4.04	3.77	-0.27	3.23	2.98	-0.25	3.48	3.22	-0.26
	(2.27)	(2.39)	(0.57)	(1.95)	(1.49)	(0.22)	(2.08)	(1.82)	(0.20)
Number of household member older than 60	0.21	0.19	-0.02	0.17	0.18	0.00	0.19	0.18	-0.00
	(0.49)	(0.40)	(0.82)	(0.44)	(0.39)	(0.96)	(0.46)	(0.39)	(0.93)
Has household member with disability/chronic disease	0.30	0.27	-0.03	0.15	0.19	0.05	0.19	0.22	0.02
	(0.46)	(0.45)	(0.77)	(0.35)	(0.40)	(0.37)	(0.39)	(0.41)	(0.60)
Number of rooms	2.89	2.27	-0.62*	1.99	1.66	-0.33*	2.26	1.84	-0.42***
	(1.24)	(1.22)	(0.01)	(1.08)	(1.05)	(0.02)	(1.20)	(1.13)	(0.00)
Household Crowding Index	3.02	3.77	0.74	3.93	4.97	1.04*	3.66	4.62	0.96**
	(1.70)	(2.62)	(0.15)	(2.87)	(3.50)	(0.02)	(2.61)	(3.30)	(0.01)
Household has impoverished roof	0.68	0.54	-0.14	0.85	0.84	-0.01	0.80	0.75	-0.05
	(0.47)	(0.51)	(0.17)	(0.36)	(0.37)	(0.86)	(0.40)	(0.44)	(0.34)
Household has impoverished floor	0.90	0.77	-0.13	0.96	0.95	-0.01	0.94	0.90	-0.05
	(0.30)	(0.43)	(0.13)	(0.19)	(0.22)	(0.69)	(0.23)	(0.30)	(0.17)
Observations	294	26	320	675	62	737	969	88	1,057

Note: Robust standard errors in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

5.2 Remote Attrition

Table 6: Attrition Table on Randomized Survey Length Assignment in Remote Survey: Completed vs. Attrition

	Own Phone			No Phone			Total Remote		
Variable	Completed	Attrition	Diff	Completed	Attrition	Diff	Completed	Attrition	Diff
List respondent age	39.03	36.09	-2.95*	39.76	39.27	-0.49	39.34	38.62	-0.72
	(12.38)	(13.01)	(0.04)	(13.94)	(14.31)	(0.73)	(13.05)	(14.11)	(0.44)
Household head is the respondent	0.78	0.68	-0.09	0.82	0.79	-0.03	0.80	0.77	-0.03
	(0.42)	(0.47)	(0.06)	(0.38)	(0.41)	(0.37)	(0.40)	(0.42)	(0.30)
Female Household Head	0.25	0.21	-0.05	0.32	0.31	-0.01	0.28	0.29	0.00
	(0.44)	(0.41)	(0.30)	(0.47)	(0.46)	(0.77)	(0.45)	(0.45)	(0.94)
Married Household Head	0.27	0.25	-0.03	0.14	0.10	-0.04	0.22	0.13	-0.08**
	(0.45)	(0.43)	(0.56)	(0.35)	(0.31)	(0.26)	(0.41)	(0.34)	(0.00)
Household head years of education	7.92	7.03	-0.89	3.54	3.26	-0.28	6.03	4.02	-2.01***
	(4.69)	(4.26)	(0.08)	(3.49)	(3.17)	(0.42)	(4.73)	(3.74)	(0.00)
Displaced household	0.15	0.08	-0.07*	0.03	0.03	-0.00	0.10	0.04	-0.06**
	(0.36)	(0.27)	(0.04)	(0.18)	(0.18)	(0.92)	(0.30)	(0.20)	(0.00)
Household size	8.39	7.50	-0.89	5.99	5.96	-0.04	7.36	6.27	-1.09***
	(3.40)	(4.42)	(0.05)	(3.05)	(2.57)	(0.91)	(3.46)	(3.10)	(0.00)
Number of household member younger than 15	4.31	3.68	-0.63**	3.13	3.13	-0.00	3.80	3.24	-0.56***
	(2.03)	(2.17)	(0.01)	(2.29)	(1.81)	(1.00)	(2.22)	(1.90)	(0.00)
Number of household member older than 60	0.21	0.24	0.03	0.13	0.19	0.05	0.18	0.20	0.02
	(0.44)	(0.64)	(0.65)	(0.41)	(0.49)	(0.21)	(0.43)	(0.53)	(0.52)
Has household member with disability/chronic disease	0.31	0.28	-0.02	0.15	0.17	0.01	0.24	0.19	-0.05
	(0.46)	(0.45)	(0.69)	(0.36)	(0.37)	(0.68)	(0.43)	(0.39)	(0.10)
Number of rooms	3.13	2.85	-0.29	1.95	1.91	-0.04	2.62	2.10	-0.53***
	(1.27)	(1.40)	(0.06)	(1.09)	(1.09)	(0.69)	(1.33)	(1.22)	(0.00)
Household Crowding Index	2.97	3.04	0.06	3.80	4.04	0.25	3.33	3.84	0.51**
	(1.26)	(1.97)	(0.74)	(2.68)	(3.02)	(0.37)	(2.04)	(2.87)	(0.00)
Household has impoverished roof	0.63	0.68	0.05	0.86	0.86	0.00	0.73	0.82	0.09**
	(0.48)	(0.47)	(0.34)	(0.35)	(0.35)	(0.99)	(0.45)	(0.38)	(0.00)
Household has impoverished floor	0.92	0.90	-0.02	0.97	0.96	-0.01	0.94	0.95	0.01
	(0.28)	(0.30)	(0.62)	(0.16)	(0.19)	(0.42)	(0.23)	(0.22)	(0.66)
Observations	157	151	308	119	596	715	276	747	1,023

Note: Robust standard errors in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

6 Primary Results: Impact of Survey Modality

In this section, we present the main results of the study aggregated by the food security indicator. Here, the objective is to compare the different versions of CARI calculated either using remote or face-to-face modules with the standard CARI.

6.1 CARI vs. rCARI

Conventional/standard CARI: CARI-FES: Standard CARI using FES from F2F surveys.

Other ways of aggregating CARI (that are tested)

- rCARI-FES: Standard CARI using FES in but coming from remote surveys.
- CARI-Inc: CARI using income variable coming from F2F surveys.
- rCARI-Inc: Standard rCARI using income and income change variable coming from remote surveys (this is the CARI version that was used during COVID 19 Pandemic).

Table 7: Impact of Survey Modality: Standard CARI-FES vs. Standard rCARI-Inc

	Own Phone			No Phone			Total		
Variable	F2F	Remote	Diff	F2F	Remote	Diff	F2F	Remote	Diff
CARI vs. rCARI: Food Secure	0.01	0.01	-0.00	0.00	0.02	0.02	0.00	0.01	0.01
	(0.08)	(0.08)	(0.96)	(0.00)	(0.13)	(0.15)	(0.05)	(0.10)	(0.17)
CARI vs. rCARI: Marginally Food Secure	0.31	0.52	0.21***	0.07	0.14	0.07*	0.14	0.36	0.22***
	(0.46)	(0.50)	(0.00)	(0.25)	(0.35)	(0.03)	(0.35)	(0.48)	(0.00)
CARI vs. rCARI: Moderately Food Insecure	0.60	0.45	-0.15**	0.74	0.70	-0.04	0.70	0.56	-0.14***
	(0.49)	(0.50)	(0.00)	(0.44)	(0.46)	(0.39)	(0.46)	(0.50)	(0.00)
CARI vs. rCARI: Severely Food Insecure	0.08	0.02	-0.06**	0.20	0.14	-0.05	0.16	0.07	-0.09***
	(0.27)	(0.14)	(0.00)	(0.40)	(0.35)	(0.14)	(0.37)	(0.26)	(0.00)
CARI vs. rCARI: Food Insecure	0.68	0.47	-0.21***	0.93	0.84	-0.09**	0.86	0.63	-0.23***
	(0.47)	(0.50)	(0.00)	(0.25)	(0.37)	(0.01)	(0.35)	(0.48)	(0.00)
Observations	294	157	451	675	119	794	969	276	1,245

Note: Robust standard errors in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

Table 8: Impact of Survey Modality: CARI-FES vs. rCARI-FES

	Own Phone			No Phone			Total		
Variable	F2F	Remote	Diff	F2F	Remote	Diff	F2F	Remote	Diff
CARI using FES: Food Secure	0.01	0.01	-0.00	0.00	0.01	0.01	0.00	0.01	0.01
	(0.08)	(0.08)	(0.96)	(0.00)	(0.09)	(0.32)	(0.05)	(0.08)	(0.33)
CARI using FES: Marginally Food Secure	0.31	0.29	-0.02	0.07	0.13	0.06	0.14	0.22	0.08**
	(0.46)	(0.45)	(0.61)	(0.25)	(0.33)	(0.07)	(0.35)	(0.41)	(0.01)
CARI using FES: Moderately Food Insecure	0.60	0.58	-0.02	0.74	0.59	-0.15**	0.70	0.58	-0.11***
	(0.49)	(0.50)	(0.65)	(0.44)	(0.49)	(0.00)	(0.46)	(0.49)	(0.00)
CARI using FES: Severely Food Insecure	0.08	0.13	0.05	0.20	0.28	0.08	0.16	0.19	0.03
	(0.27)	(0.33)	(0.14)	(0.40)	(0.45)	(0.06)	(0.37)	(0.39)	(0.24)
CARI using FES: Food Insecure	0.68	0.71	0.02	0.93	0.87	-0.07*	0.86	0.78	-0.08**
	(0.47)	(0.46)	(0.61)	(0.25)	(0.34)	(0.04)	(0.35)	(0.42)	(0.00)
Observations	294	157	451	675	119	794	969	276	1,245

Note: Robust standard errors in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

Table 9: Impact of Survey Modality: CARI-Inc vs. rCARI-Inc

	Own Phone			No Phone			Total		
Variable	F2F	Remote	Diff	F2F	Remote	Diff	F2F	Remote	Diff
rCARI using Income: Food Secure	0.02	0.01	-0.01	0.00	0.02	0.02	0.01	0.01	0.00
	(0.14)	(0.08)	(0.18)	(0.04)	(0.13)	(0.20)	(0.08)	(0.10)	(0.59)
rCARI using Income: Marginally Food Secure	0.51	0.52	0.01	0.24	0.14	-0.09**	0.32	0.36	0.04
	(0.50)	(0.50)	(0.86)	(0.43)	(0.35)	(0.01)	(0.47)	(0.48)	(0.25)
rCARI using Income: Moderately Food Insecure	0.45	0.45	0.01	0.72	0.70	-0.03	0.64	0.56	-0.08*
	(0.50)	(0.50)	(0.89)	(0.45)	(0.46)	(0.55)	(0.48)	(0.50)	(0.02)
rCARI using Income: Severely Food Insecure	0.02	0.02	-0.00	0.04	0.14	0.11**	0.03	0.07	0.04*
	(0.14)	(0.14)	(0.92)	(0.19)	(0.35)	(0.00)	(0.18)	(0.26)	(0.01)
rCARI using Income: Food Insecure	0.47	0.47	0.01	0.76	0.84	0.08*	0.67	0.63	-0.04
	(0.50)	(0.50)	(0.91)	(0.43)	(0.37)	(0.04)	(0.47)	(0.48)	(0.21)
Observations	294	157	451	675	119	794	969	276	1,245

Note: Robust standard errors in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

6.2 FCS

Table 10: Impact of Survey Modality on Food Consumption Score and Groups

	Own Phone			No Phone			Total		
Variable	F2F	Remote	Diff	F2F	Remote	Diff	F2F	Remote	Diff
Food Consumption Score	39.76	47.63	7.87***	24.59	28.12	3.53**	29.19	39.22	10.03***
	(17.58)	(17.45)	(0.00)	(12.67)	(12.00)	(0.00)	(15.94)	(18.12)	(0.00)
FCS: Acceptable	0.40	0.58	0.18***	0.10	0.09	-0.01	0.19	0.37	0.18***
	(0.49)	(0.50)	(0.00)	(0.31)	(0.29)	(0.70)	(0.40)	(0.48)	(0.00)
FCS: Borderline	0.35	0.29	-0.06	0.24	0.39	0.15**	0.27	0.33	0.06
	(0.48)	(0.45)	(0.19)	(0.43)	(0.49)	(0.00)	(0.45)	(0.47)	(0.07)
FCS: Poor	0.25	0.13	-0.12**	0.66	0.52	-0.14**	0.53	0.30	-0.23***
	(0.43)	(0.34)	(0.00)	(0.48)	(0.50)	(0.01)	(0.50)	(0.46)	(0.00)
Consumption over the past 7 days (cereals and tubers)	5.87	6.24	0.36*	5.08	4.87	-0.21	5.32	5.64	0.33*
	(1.86)	(1.33)	(0.02)	(2.17)	(2.13)	(0.31)	(2.11)	(1.85)	(0.01)
Consumption over the past 7 days (pulses)	3.74	3.74	-0.01	2.01	1.98	-0.03	2.54	2.98	0.45**
	(2.33)	(2.08)	(0.98)	(1.92)	(1.62)	(0.88)	(2.20)	(2.08)	(0.00)
Consumption over the past 7 days (dairy products)	0.50	1.14	0.64**	0.07	0.40	0.33**	0.20	0.82	0.62***
	(1.57)	(2.27)	(0.00)	(0.55)	(1.11)	(0.00)	(0.99)	(1.90)	(0.00)
Consumption over the past 7 days (protein-rich foods)	1.72	2.09	0.37*	0.51	0.59	0.08	0.88	1.44	0.56***
	(1.81)	(1.81)	(0.04)	(1.07)	(0.99)	(0.45)	(1.45)	(1.68)	(0.00)
Consumption over the past 7 days (vegetables)	3.73	5.64	1.92***	4.11	5.56	1.46***	3.99	5.61	1.62***
	(2.93)	(1.66)	(0.00)	(2.75)	(1.78)	(0.00)	(2.81)	(1.71)	(0.00)
Consumption over the past 7 days (fruit)	0.54	0.72	0.18	0.31	0.59	0.27*	0.38	0.66	0.28**
	(1.31)	(1.34)	(0.17)	(1.03)	(1.22)	(0.02)	(1.13)	(1.29)	(0.00)
Consumption over the past 7 days (oil)	4.90	5.79	0.89***	1.99	2.78	0.79***	2.87	4.49	1.62***
	(2.37)	(1.81)	(0.00)	(2.38)	(2.24)	(0.00)	(2.73)	(2.50)	(0.00)
Consumption over the past 7 days (sugar)	2.36	3.53	1.17***	1.30	1.86	0.55**	1.62	2.81	1.18***
	(2.60)	(2.78)	(0.00)	(2.02)	(1.98)	(0.01)	(2.26)	(2.60)	(0.00)
Observations	294	157	451	675	119	794	969	276	1,245

Note: Robust standard errors in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

6.3 rCSI

Table 11: Impact of Survey Modality on Reduced Coping Strategy Index Classification

	Own Phone			No Phone			Total		
Variable	F2F	Remote	Diff	F2F	Remote	Diff	F2F	Remote	Diff
Reduced Coping Strategies Index (rCSI)	14.35	17.27	2.92**	13.16	16.75	3.59***	13.52	17.05	3.52***
	(9.97)	(8.88)	(0.00)	(10.70)	(9.82)	(0.00)	(10.49)	(9.28)	(0.00)
rCSI: Low	0.06	0.04	-0.02	0.19	0.07	-0.13***	0.15	0.05	-0.10***
	(0.25)	(0.21)	(0.36)	(0.40)	(0.25)	(0.00)	(0.36)	(0.23)	(0.00)
rCSI: Median	0.70	0.57	-0.14**	0.55	0.57	0.02	0.60	0.57	-0.03
	(0.46)	(0.50)	(0.00)	(0.50)	(0.50)	(0.66)	(0.49)	(0.50)	(0.41)
rCSI: Severe	0.23	0.39	0.16***	0.26	0.36	0.11*	0.25	0.38	0.13***
	(0.42)	(0.49)	(0.00)	(0.44)	(0.48)	(0.03)	(0.43)	(0.49)	(0.00)
Relied on less preferred, less expensive food	3.92	2.75	-1.17***	3.39	3.29	-0.10	3.55	2.98	-0.57***
	(2.00)	(1.84)	(0.00)	(2.43)	(2.45)	(0.67)	(2.32)	(2.14)	(0.00)
Borrowed food or relied on help from friends or relatives	0.77	0.89	0.12	0.82	0.74	-0.08	0.81	0.82	0.02
	(1.15)	(1.25)	(0.32)	(1.26)	(1.03)	(0.43)	(1.23)	(1.16)	(0.84)
Reduced portion size of meals at meals time	1.98	2.65	0.67**	2.01	2.45	0.44*	2.00	2.57	0.57***
	(2.27)	(2.14)	(0.00)	(2.39)	(2.09)	(0.04)	(2.35)	(2.12)	(0.00)
Restricted consumption by adults in order for young-children to eat	1.73	2.70	0.97***	1.52	2.42	0.90***	1.58	2.58	1.00***
	(1.90)	(1.83)	(0.00)	(1.80)	(2.16)	(0.00)	(1.83)	(1.98)	(0.00)
Reduced the number of meals eaten per day	1.74	2.00	0.26	1.55	2.27	0.72***	1.61	2.12	0.51***
	(2.38)	(2.12)	(0.24)	(2.42)	(2.06)	(0.00)	(2.41)	(2.09)	(0.00)
Observations	294	157	451	675	119	794	969	276	1,245

Note: Robust standard errors in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

6.4 LCS

Table 12: Impact of Survey Modality on Livelihood Coping Strategy Index

	Own Phone			No Phone			Total		
Variable	F2F	Remote	Diff	F2F	Remote	Diff	F2F	Remote	Diff
Maximum Livelihood Coping: None	0.20	0.11	-0.09*	0.39	0.13	-0.26***	0.33	0.12	-0.21***
	(0.40)	(0.32)	(0.01)	(0.49)	(0.34)	(0.00)	(0.47)	(0.33)	(0.00)
Maximum Livelihood Coping: Stress level	0.61	0.43	-0.19***	0.43	0.39	-0.04	0.48	0.41	-0.07*
	(0.49)	(0.50)	(0.00)	(0.50)	(0.49)	(0.39)	(0.50)	(0.49)	(0.03)
Maximum Livelihood Coping: Crisis level	0.14	0.26	0.13**	0.13	0.21	0.08*	0.13	0.24	0.11***
	(0.34)	(0.44)	(0.00)	(0.33)	(0.41)	(0.04)	(0.34)	(0.43)	(0.00)
Maximum Livelihood Coping: Emergency level	0.05	0.20	0.15***	0.05	0.27	0.22***	0.05	0.23	0.18***
	(0.22)	(0.40)	(0.00)	(0.22)	(0.45)	(0.00)	(0.22)	(0.42)	(0.00)
Observations	294	157	451	675	119	794	969	276	1,245

Note: Robust standard errors in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

Table 13: CAR: Each Coping Strategy Used

	Own Phone			No Phone			Total		
Variable	F2F	Remote	Diff	F2F	Remote	Diff	F2F	Remote	Diff
Used Stress coping: Sold household assets/goods	0.28	0.46	0.18***	0.15	0.29	0.14**	0.19	0.38	0.20***
	(0.45)	(0.50)	(0.00)	(0.35)	(0.45)	(0.00)	(0.39)	(0.49)	(0.00)
Used Stress coping: Reduced expenses on health and education	0.23	0.43	0.20***	0.18	0.40	0.23***	0.19	0.42	0.22***
	(0.42)	(0.50)	(0.00)	(0.38)	(0.49)	(0.00)	(0.39)	(0.49)	(0.00)
Used Stress coping: Spent savings	0.47	0.50	0.03	0.22	0.31	0.09	0.30	0.42	0.12***
	(0.50)	(0.50)	(0.50)	(0.42)	(0.46)	(0.06)	(0.46)	(0.49)	(0.00)
Used Stress coping: Borrowed money	0.46	0.53	0.07	0.31	0.34	0.02	0.36	0.45	0.09**
	(0.50)	(0.50)	(0.18)	(0.46)	(0.47)	(0.62)	(0.48)	(0.50)	(0.01)
Used Crisis coping: Sold productive assets or means of transport	0.07	0.26	0.19***	0.02	0.07	0.05*	0.04	0.18	0.14***
	(0.26)	(0.44)	(0.00)	(0.14)	(0.25)	(0.05)	(0.19)	(0.38)	(0.00)
Used Crisis coping: Moved to somewhere else	0.05	0.14	0.09**	0.11	0.24	0.13**	0.09	0.18	0.09***
	(0.23)	(0.35)	(0.01)	(0.31)	(0.43)	(0.00)	(0.29)	(0.39)	(0.00)
Used Crisis coping: Withdrew children from school	0.03	0.05	0.02	0.03	0.09	0.06*	0.03	0.07	0.04*
	(0.17)	(0.22)	(0.32)	(0.17)	(0.29)	(0.02)	(0.17)	(0.25)	(0.01)
Used Emergency coping: Sold houses	0.03	0.14	0.11***	0.02	0.22	0.20***	0.02	0.17	0.15***
	(0.17)	(0.35)	(0.00)	(0.15)	(0.41)	(0.00)	(0.16)	(0.38)	(0.00)
Used Emergency coping: Begged and/or scavenged	0.01	0.07	0.06**	0.02	0.08	0.05*	0.02	0.07	0.05**
	(0.12)	(0.26)	(0.01)	(0.15)	(0.27)	(0.04)	(0.14)	(0.26)	(0.00)
Used Emergency coping: Engaged in illegal income activities	0.01	0.03	0.02	0.01	0.05	0.04*	0.01	0.04	0.03*
	(0.10)	(0.16)	(0.27)	(0.09)	(0.22)	(0.04)	(0.10)	(0.19)	(0.02)
Observations	294	157	451	675	119	794	969	276	1,245

Note: Robust standard errors in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

6.5 FES

Table 14: Impact of Survey Modality on Expenditure and Food Expenditure Share

	Own Phone			No Phone			Total		
Variable	F2F	Remote	Diff	F2F	Remote	Diff	F2F	Remote	Diff
Food Expenditure Share	0.67	0.73	0.06*	0.73	0.68	-0.06*	0.71	0.70	-0.01
	(0.20)	(0.24)	(0.01)	(0.17)	(0.25)	(0.02)	(0.18)	(0.24)	(0.52)
FES < 50 percent	0.20	0.16	-0.04	0.10	0.25	0.15***	0.13	0.20	0.07*
	(0.40)	(0.37)	(0.27)	(0.30)	(0.44)	(0.00)	(0.34)	(0.40)	(0.01)
FES: 50-65 percent	0.23	0.17	-0.06	0.16	0.18	0.02	0.18	0.17	-0.00
	(0.42)	(0.38)	(0.15)	(0.36)	(0.38)	(0.61)	(0.38)	(0.38)	(0.86)
FES: 65-75 percent	0.13	0.13	-0.01	0.17	0.11	-0.06	0.16	0.12	-0.04
	(0.34)	(0.33)	(0.87)	(0.38)	(0.31)	(0.06)	(0.37)	(0.33)	(0.08)
FES: >= 75 percent	0.44	0.54	0.10*	0.57	0.46	-0.11*	0.53	0.51	-0.02
	(0.50)	(0.50)	(0.04)	(0.50)	(0.50)	(0.03)	(0.50)	(0.50)	(0.50)
Observations	294	157	451	675	119	794	969	276	1,245

Note: Robust standard errors in parentheses; * p<0.05, ** p<0.01, *** p<0.001.
In Myanmar, the Short Form in Remote Survey was dropped due to translation issues.

6.6 Income

Table 15: Impact of Survey Modality on Income

	Own Phone			No Phone			Total		
Variable	F2F	Remote	Diff	F2F	Remote	Diff	F2F	Remote	Diff
Income: Regular and Unchanged/Increasing	0.45	0.26	-0.19***	0.44	0.24	-0.19***	0.44	0.25	-0.19***
	(0.50)	(0.44)	(0.00)	(0.50)	(0.43)	(0.00)	(0.50)	(0.44)	(0.00)
Income: Regular and Reduced OR Irregular and Unchanged/Increasing	0.42	0.50	0.08	0.46	0.49	0.02	0.45	0.49	0.04
	(0.49)	(0.50)	(0.13)	(0.50)	(0.50)	(0.63)	(0.50)	(0.50)	(0.22)
Income: Irregular and Reduced	0.12	0.23	0.11**	0.08	0.25	0.17***	0.09	0.24	0.15***
	(0.32)	(0.42)	(0.00)	(0.27)	(0.44)	(0.00)	(0.29)	(0.43)	(0.00)
Income: No income sources	0.01	0.01	-0.00	0.02	0.02	-0.00	0.02	0.01	-0.00
	(0.12)	(0.11)	(0.94)	(0.14)	(0.13)	(0.85)	(0.13)	(0.12)	(0.71)
Observations	294	157	451	675	119	794	969	276	1,245

Note: Robust standard errors in parentheses; * p<0.05, ** p<0.01, *** p<0.001.